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An Experimental Investigation on Unsupervised Text Summarization for Customer Reviews

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Abstract

People generally turn to websites such as Yelp to find reviews for the food/restaurant before trying it first-hand. However, some reviews are so long and ambiguous that users get confused about whether the review appreciates or disparages the food. Here, websites might want to employ summarization techniques so that the crux of the study is well understood in just a sentence. The process of compressing the information/data while preserving its idea is text summarization. The present work's main objective is to empirically investigate the suitable algorithm for summarizing reviews so that users do not need to spend time reading through the entire study. The performance of the conducted investigations is analyzed with the metrics such as cosine similarity, the score of the bilingual evaluation understudy (precision), and recall-oriented understudy for gisting evaluation analysis. The experiments show that the text summarization technique, LexRank outperforms other methods with the precision and recall values of 0.586 and 0.346, respectively.

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1. Introduction

The data inundation in the world has been a significant reason for the emphasis and accomplishments of text [1]. This overwhelmingly much information gives rise to various sophisticated and efficient text summarizers. Even before the advancement of technology, text summarization has profound applications – from abstracts of journals, scientific

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papers, and news articles to summaries of books, digests, etc. This growth of demand for an enhanced method of text summarization with minimum utilization of time has been a significant subject of research in many of the finest universities across the globe, like the Laboratory of Topic Maps at Leipzig University, Germany, Concordia University in Montreal, Canada, the Faculty of Informatics in Masaryk University, Czech Republic, and the IHR Nexus Lab at Arizona State University at Arizona, USA [2].

The concept of text summarization has entrenched deeply as a pivotal engine or tool to support and illustrate vast amounts of texts from the swiftly emergent information age [3]. In reality, humongous documents are far more complex for ordinary humans to summarize their entire content manually [4]. On top of that, there is also a plethora of information on the internet above and beyond what is desired. For this reason, several paradigms have also been developed for searching, processing, retrieving, and storing data [5]. The problem becomes two-fold: Looking for appropriate documents/information from the mind-boggling number of reports offered and stupendously high volume of salient details. Hence, the main object of text summarization is to compress the original text into an accurate version of the text while preserving its context and global denotation. The primary virtue of text summarization is the reduction in the reading time of the user. An excellent text summarization tool must emulate the document's theme by keeping the redundancy minimum. The methods of summarization are restricted to extractive and abstractive summarization.

In extractive text summarization, a subset of seminal existing words, sentences, or phrases is extracted directly from the original text [6]. Extractive summarization can be distinguished from abstractive summarization, which complements its extractive coequal. It aims to synthesize new phrases or sentences from scratch rather than extract them from the original set of the report(s). In more straightforward thoughts, extractive summarization can be considered a highlighter pen, while abstractive summarization can be declared a human writer [7]. Abstractive approaches are harder to develop because effective and efficient natural language generation methodologies are required, which is an active research field [8].

The main objective of the present work is to investigate the suitable algorithm empirically for summarizing reviews so that the users need not spend time reading through the entire study. The performance of the conducted investigations is analyzed with the metrics such as cosine similarity, the score of the bilingual evaluation understudy (precision), and recall-oriented understudy for gisting evaluation analysis.

The organization of this paper is as follows, and section 2 is explained with surveys on the existing research whose inferences are aligned with our work. Section 3 is briefed on the methodology followed during the experiment and explains the dataset features and preprocessing techniques employed. Section 4 describes each algorithm used for the comparative study and evaluation of different methods. The results obtained from the deep learning model are discussed in section 5. The concluding remarks are presented in section 6.

2. Literature Survey

The entire gamut of text summarization can be broadly categorized based on the input, the purpose of summarization, and the type of output produced by the summarizer (illustrated in Fig. 1.). This study currently confines itself to unsupervised extractive type text summarization.

Generally, text summarization involves preprocessing, feature extraction, and summarization. Commonly used preprocessing methodologies include but are not limited to case folding (converting every character to lowercase), removal of stop words (like the, a, an, etc.), and stemming (conversion of words to their corresponding stems), parts of speech tagging. In the feature extraction step, the sentences are then analyzed based on either statistic (emphasis on the mathematical representations) or linguistic (emphasis on the actual meaning/context of the sentence) features (sometimes hybrid analysis involving both statistical and linguistic features is used). An algorithm is then applied to the extracted features to get the summary structure from the text structure wherein the sentences are ranked. In the generation step, the sentences are ranked, and the essential sentences/phrases are picked up to generate a summary.

The techniques used for extractive text summarization are reviewed by emphasizing unsupervised approaches [9]. The authors implemented various statistical approaches, graphical, machine learning, latent semantic analysis, and fuzzy logic approaches. The statistical approaches proved simple and fast without a dataset for training purposes. Graph-based approaches can generate topic-specific or query-specific summaries with better information coverage. Machine-learning approaches can inappropriately represent document features, test the performance of many features, and provide a solution for sentence scoring problems. Latent semantic approaches are the best for semantic relations and generate good coverage knowledge with minor noise, but they still suffer from polysemy problems. Fuzzy logic approaches are an excellent alternative for improving sentence scoring problems and enhancing summarization if

integrated with other techniques, but human experts must define fuzzy rules. A convolutional neural network for extractive text summarization is developed in [10]. The document understanding conference dataset was used to train the model. The word embeddings of the sentences were obtained using Google's word2vec.

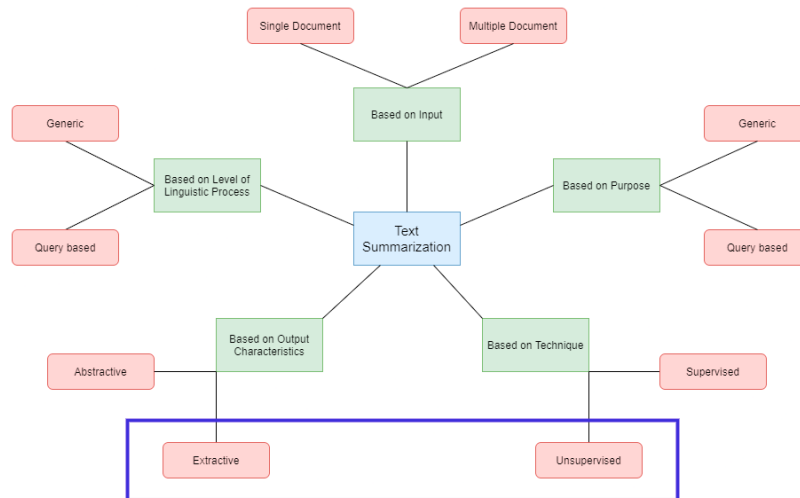


Fig. 1. Taxonomy of existing text summarizers.

The exploration in [11] proposed extractive text summarization using features like the frequency of words, position, skeleton, and cue words by manually allocating the weights for every feature. Over 400 documents that were manually generated were used for testing. The algorithm produced exceptional results, with around 44% of the summaries generated by the system matching the manual summaries. Sun et al. [12] suggested a method of automatic text summarization using reinforcement learning. Here, the research was also undertaken to summarize the articles from wikipedia. The various techniques for summarizing the wikipedia pages are effectuated in [13]. In their most successful method, the words were stemmed, and the word frequency of the root forms was considered to assign equal weights to words with the same root. Also, the sentences with citations were given a higher weightage than ordinary sentences.

Mishra et al. [14] presented a summarization technique for scientific documents that requires extracting figures, structure, and tables, identifying pertinent entities, and generating a helpful summary. The summarization module of the proposed system seeks to generate compact yet comprehensive summaries in the process. The approach was validated with the help of human experts. An iteration-based extractive text summarizer [15]. People often read an article more than once to comprehend and fully summarize its contents. This line of thought heavily influenced the approach used, wherein the extent to which a selective reading mechanism determined each phrase in the model. The final labelling part uses outputs from all iterations to generate summaries.

Bidirectional encoder representations from transformers (BERT) are an alternative language representation model [16]. In BERT, the representations of unlabeled text are conditioned by the right and left context in each layer, unlike other language representation models. The pre-trained BERT is generally finetuned with an additional refinement layer for producing the outputs of various tasks such as language inference question answering without making separate task-specific alterations to the model's architecture. When BERT was compared against previously benchmarked models, it obtained better results in many natural language processing (NLP) tasks or applications.

A common framework for extractive and abstractive summarization while depicting BERT's applications is investigated with deep learning-based abstractive text summarization [17]. They propose a new BERT-based document-level encoder that can obtain representations for its sentences and express the semantics of the documents. Several inter sentence transformer layers are stacked on top of each other as an extractive model that wraps the BERT-

based encoder. For abstractive text summarization, various optimizers for the encoder-decoder were proposed to lessen the mismatch between the two (the former is pre-trained while the latter is not).

A simple variation of BERT is called BERTSUM for extractive text summarization [18]. Here, several summarization layers that can be applied with BERT have been experimented with on a trial basis. The system (specifically BERTSUM + Transformer) used the CNN/Dailymail dataset and outperformed the previous best-performed system by 1.65 on recall-oriented understudy for gisting evaluation (ROUGE).

Madhuri et al. [19] proposed an extractive text summarization mechanism wherein the sentences are ranked before summarizing. The sentences from the input document are initially tokenized, followed by general preprocessing steps like stop word removal. Then the introductory sentences in the document are ranked (using a weighted frequency of the words) higher than the other sentences. These meaningful sentences are then extracted, resulting in a high-quality summary of the input document. Vijay et al. [20] built a text summarizer capable of summarizing news articles across different categories like World, Science & Technology, Sports, etc., in Hindi. Due to the unavailability of Hindi datasets, 24,253 news articles were extracted from inshorts.com with the manual gold summaries, each containing exactly 60 words. Various word level and sentence level features were considered for the summarization task. A text summarizer using the Essex Arabic summaries corpus and a manual summary [21]. This model was assessed using the ROUGE. The work has outperformed two other extractive Arabic summarizers.

Gidiotis et al. [22] proposed a novel text summarizer using divide and conquer-based neural summarization. The whole idea is to divide the main task into multiple subtasks. In this case, a document is divided into multiple key-value pairs and its summary. This methodology increases the number of training examples, resulting in increased performance. The fragmentary summaries are then integrated to form the decisive summary of the input. This method also reduces the computational complexity. Arman et al. [23] argued that ROUGE is not practical in variations in terminology and paraphrasing since ROUGE relies on lexical overlaps between phrases and terms of a sentence. Here, it provides an extensive analysis of the effectiveness of the ROUGE set of metrics and proposes a novel metric titled summarization evaluation by relevance analysis that achieves a high correlation with manual scores indicating effective article summarization.

The investigation in [24] analyses the various types and methods of text summarization after outlining the significant steps required for any standard text summarizer – topic identification, interpretation, and summary generation. A survey of extractive text summarization performed by Joshi et al. [25] reviewed benchmarked datasets and challenges of extractive text summarization. Here, the various features of words and sentences have also been briefly explored. The importance of feature weights in choosing key sentences has been shown. Vetriselvi et al. [26] provide a systematic and broad overview of research in text summarization. Various recommendations to address the challenges and future opportunities are also discussed. The exploration and investigation in the research work [27] focus on a systematic analysis of various text summarization techniques after introducing concepts related to extractive text summarization. Problems with similarity measurement and textual evaluation have also been significantly emphasized.

3. Methodology

An in-depth literature survey of pre-existing works related to text summarization using various NLP techniques gathered information on various popular tools, technologies, and mechanisms. The Amazon fine foods review dataset data is processed using standard preprocessing techniques to clean the data, such as removing stop words, expanding contractions, etc. The cleaned data is reduced to its word stem using the inbuilt packages subjected to nine different algorithms - LexRank, Luhn, LSA, TextRank, Edmundson, Sum Basic, KL, Reduction & PageRank to generate summaries [28]. The results were analyzed using cosine similarity, ROUGE, and bilingual evaluation understudy (BLEU) scores. The reviews are processed with various tools, and preprocessing methods are used to perform on cleaning the data. After preprocessing, the features are extracted and then applied with any one of the selected algorithms. The summary is generated as the output of the algorithm. Then, ROUGE and BLEU scores are computed to compare the final results. The methodology is depicted in Fig. 2.

tea cat much coffee food packaging best
beware cookie look something's waste water nasty little small quality
way baby hit taste really eat got high new use can't dog
better bitter wrong bland made shipped terrible worst
bad taste plastic flavor poor sugar organic weak yuck
product sugar organic weak yuck
amazon batch sweet bag need buy stale hot corn syrup gross
great n't good disappointing
love chip horrible disappointed contains
favorite salt work taste awful
old under took
leading
love



Sentiment	count
positive	450000
negative	80000

Box plot showing the distribution of text_word_count for scores 1 through 5. The y-axis is labeled 'text_word_count' and ranges from 0 to 250. The x-axis is labeled 'Score' and has categories 1, 2, 3, 4, and 5. Each score has a corresponding box plot: Score 1 is blue, Score 2 is orange, Score 3 is green, Score 4 is red, and Score 5 is purple. The median text_word_count increases from Score 1 to Score 3 and then decreases for Scores 4 and 5. The interquartile range (IQR) is largest for Score 3.

Fig. 7. Average review length.

A flask-based web application was created during this study to compare the summarization of texts from various methods. This web-based comparison involved extractive and abstractive summarization techniques provided by more than one tool (NLTK, Sumy, Gensim, etc.).

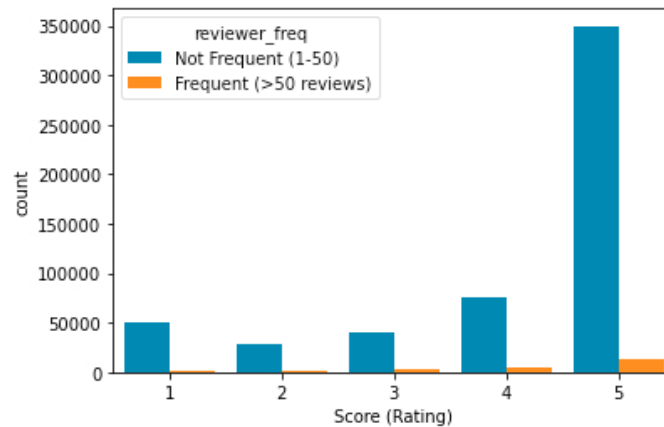


Fig. 8. Reviewer frequency.

4. Implementation

The data in the raw form contains user id, helpfulness numerator & denominator, product id, profile name, the score or ratings, submission time, summary, and the text. The dataset also contains null values. As the initial step, because of over 1 million data entries, features other than the actual review and summary of the review were dropped along with the tuples containing null values for reviews or summaries. Contractions in each review and summary were expanded. Each set of reviews and summary was cleaned by removing stop words and other undesired characters to get two lists containing lists of reviews and their corresponding summaries. Before proceeding with the implementation, basic preprocessing steps like dropping not a number (NaN) values and removing stop words were performed. After this dataset, a comparative study was made between nine popular unsupervised algorithms (LexRank, LUHN, LSA, TextRank, Edmundson, SumBasic, KL, Reduction, Pagerank) used for text summarization. After creating the summaries, three evaluation metrics are used to compare the models, i.e., Cosine similarity, BLEU (Precision), and ROUGE for all nine algorithms, and their descriptions are discussed.

LexRank is an unsupervised text summarization algorithm using graph-based centrality to score sentences. The LexRank algorithm works by recommending sentences. Thus, if a sentence or phrase is comparable to other sentences in the original text, it is regarded as a sentence of great importance. Therefore, it must have more than many sentences in the original text for a sentence to be present in the final summary. The main idea is that sentences “recommend” other similar sentences to the reader. Luhn’s algorithm is based on the concept of Term Frequency – Inverse Document Frequency. In the given text, a word is of higher importance if the frequency of the word is high, i.e., the word occurs repeatedly. The words at the beginning of the sentences are assigned higher weights.

Latent semantic index (LSI) or Latent semantic analysis (LSA) uses a bag of word (BoW) model that produces a term-document matrix (denoting the number of occurrences of each term/word in a document). The rows and columns represent terms and documents, respectively. LSA/LSI uses singular value decomposition to perform matrix decomposition on the term-document matrix to learn the latest topics. TextRank is a graph-based ranking model for processing text to find relevant sentences and keywords. The individual sentences are transformed into vectors. Then the distance between these vectors is stored in a similarity matrix, then transformed into a graph with sentences as nodes and the similarity scores as the edge weights. The sentences with higher ranks are displayed as summaries.

Edmundson’s algorithm can be considered an extension or enhancement of Luhn’s algorithm. This algorithm is language-dependent as it requires “bonus words”, words stated by the user as of high importance (positively relevant),

“stigma words”, words of low importance or even negative importance (negatively relevant); and “stop words” or “null words” (irrelevant words), which are the same as used elsewhere in NLP processing.

SumBasic algorithm is generally used to generate multi-document text summaries. The algorithm uses the word probability approach to determine the importance of a sentence, i.e., utilize the words that occur more frequently in a document over the words with less frequency to generate a more likely human abstract summary. The KLSum algorithm is based on word distribution similarity. It selects the sentences or phrases in a greedy approach that includes the sentences or phrases in summary as long as the KL Divergence (to measure the differences between two or more probability distributions over a single variable, i.e., an entropy measure) is decreasing. The Reduction algorithm is used for text summarization by using a graph-based model (where the nodes are the sentences, and the edges are the weights assigned) that evaluates the sentences in the input for the sum of the weights of their edge’s other sentences. The weights are calculated the same way as those in the TextRank model. The PageRank algorithm is popular as used by search engines such as Google to rank web pages. The PageRank algorithm has profound applications in the web’s rapid growth in size and demand for greater flexibility. This algorithm is modified to accommodate sentences instead of web pages to produce summaries for the input text.

The summaries generated from each algorithm were compared using three metrics: cosine similarity, BLEU, and ROUGE. Finally, Flask’s website is built to perform text summarization using different algorithms. The website can produce summaries of texts that the user inputs and summaries from various websites given the URL of the website. An approximate reading time is also calculated and displayed for the original and summary text.

5. Performance Analysis and Results

The nine algorithms for text summarization generated the summaries. They were compared against the golden summary provided in the amazon fine foods review dataset using cosine similarity, ROUGE score, and BLEU metric. Cosine similarity measures the similarity between two documents that are subjected to review. The maximum possible score is one, while the minimum possible score is zero, with a higher score implying better results (higher similarity). The cosine similarity analysis is listed in Table 1.

Table 1. Cosine similarity analysis.

S.No	Algorithm	Cosine similarity
1	LexRank	0.2934
2	Luhn	0.2902
3	LSA	0.1530
4	TextRank	0.2180
5	Edmundson	0.1847
6	Sum Basic	0.1276
7	KL	0.1742
8	Reduction	0.1742
9	PageRank	0.0663

ROUGE is a set of metrics that are used in various NLP tasks. For text summarization, the widely used metric is ROUGE-L, where L stands for longest common subsequence. ROUGE-L works on the simple idea that the longest common subsequence signifies the most similarity. The maximum possible score is one, while the minimum possible score is 0, with a higher score implying better results (higher similarity), and ROUGE analysis is listed in Table 3. It is evident that even though KL summarizer and reduction summarizer performs well in terms of BLEU score and Luhn summarizer also has a better result in terms of ROUGE score, and the LexRank algorithm dominates over the other eight algorithms when considering all the three metrics used to evaluate as shown in Fig. 9.

Table 2. BLEU analysis.

S.No	Algorithm	BLEU
1	LexRank	0.586
2	Luhn	0.573
3	LSA	0.273
4	TextRank	0.482
5	Edmundson	0.222
6	Sum Basic	0.385
7	KL	0.586
8	Reduction	0.586
9	PageRank	0.141

Table 3. ROUGE analysis.

S.No	Algorithm	ROUGE
1	LexRank	0.346
2	Luhn	0.346
3	LSA	0.196
4	TextRank	0.000
5	Edmundson	0.246
6	Sum Basic	0.183
7	KL	0.196
8	Reduction	0.196
9	PageRank	0.183

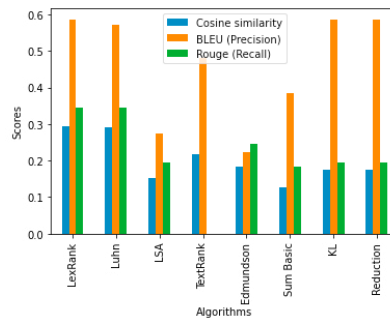


Fig. 9. Comparative analysis.

6. Conclusion

The paper presents an empirical analysis of various unsupervised extraction-based text summarising techniques. The paper thoroughly explains the currently used text summarizers and their uses, particularly in the case of restaurant or food reviews. The Amazon Fine Foods Review dataset's comparative analysis of nine well-known unsupervised extractive text summarization techniques reveals that the LexRank method outperforms the others in terms of performance when cosine similarity and the BLEU ROUGE score are taken into account. Contrary to most other algorithms (including TextRank), the LexRank algorithm can be used to summarise several documents. Taking into account the placement and length of the phrases, it also assists in maintaining redundancy and enhancing coherence.

Although text summarising has advanced significantly since its inception, effective methods that produce summaries comparable to those produced by humans and metrics that can quickly assess the summary are still lacking. The target and the input to train the model are typically required by text summarizers that can provide accurate summaries, which may not be perfect in real-time. Text summarization appears to have greater promise for machine learning-based algorithms, such as deep neural networks and contemporary reinforcement learning, which can be expanded for further study.

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